

Integrating Residual and Spatial Attention in Invertible Networks for Image Steganography

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Abstract—Image steganography seeks to conceal confidential data in an unnoticeable way through cover images for covert transmission. However, most existing methods are unable to maintain good visual effects or robust performance for Images containing Complex Textures. In light of these deficiencies, we have introduced the Residual-Spatial Attention Invertible Neural Network (RSAINN) to integrate robust feature representation and attended-learning. It can perceive the faint details on objects and among other different levels of information, thereby adjusting its focusing point as needed. Residual-Spatial Attention Module fuses multi-level low-level features hierarchically through hierarchical skip connections, improving feature fusion and avoiding severe visual distortions for enhancing stego-image security. Also improve the refined network structure through modifications to convolutional layers or other means; deepen their application to enhance feature refinement capabilities. Based on experimental results for DIV2K, COCO and ImageNet, we have shown that our method can significantly improve the stego-image Quality; The increase in PSNR is 0.35dB, 6.61dB and 4.8db than baselines respectively.

Index Terms—Image steganography, Residual-spatial attention, Skip connection, Invertible neural networks

I. INTRODUCTION

Since its initial exploration in the 1940s, steganographic techniques for digital media have transitioned into reliable tools for covert messaging in commercial and strategic environments. Given the dominance of visual data in the current internet landscape, images now serve as a ubiquitous carrier for maintaining data confidentiality. This shift amplifies concerns about data security and prompts clear terminology: a cover image is the original, unaltered host before embedding, while a stego image is the visually indistinguishable result that contains the hidden payload after embedding.

Conventional steganography typically operates in either the spatial or transform (frequency) domain. Techniques localized in the spatial domain achieve data embedding by making immediate adjustments to pixel intensities; prominent examples include difference expansion, histogram shifting, and Least Significant Bit (LSB) replacement [1]. While simple and efficient, these approaches tend to be fragile: under common perturbations (e.g., compression, noise, filtering), the embedded data may be lost, and modern steganalysis tools can detect such manipulations with high accuracy. Transform-domain

methods instead alter frequency coefficients after mapping the image from the spatial to a transform domain and then embed the secret there. This tends to improve robustness in some scenarios but often comes with constrained capacity; pushing capacity can noticeably degrade visual quality.

Modern breakthroughs in the domain of deep learning have facilitated the concealment of significantly more expansive and multifaceted information [2]. GANs have been a popular choice for this pipeline. However, their inherent training instability—stemming from the adversarial dynamics—can impede convergence, leading to fluctuating stego quality and unreliable payload preservation [3]. In recent years, the adoption of invertible neural networks (INNs) has facilitated secret image concealment characterized by precise recovery, a direct result of the inherent mathematical reversibility between the embedding and extraction stages [4].

Despite these gains, INN-based methods still encounter challenges on images with complex textures. Arranging invertible blocks purely in sequence can yield weak interconnections, causing information decay and texture-copy artifacts in generated stego images. The defensive capability against steganalysis remains inadequate, and the failure to fully preserve deep feature characteristics from both cover and secret images often degrades the accuracy of the extracted payload. These issues highlight the need for architectures that strengthen feature interaction, enhance robustness, and better preserve high-level semantics during the embedding–extraction cycle.

To overcome the aforementioned limitations, we develop the Residual-Spatial Attention Invertible Neural Network (RSAINN), a reversible steganographic architecture that integrates residual with spatial attention mechanisms. The main results of the research work are listed below:

- 1) We present a proposed RSAINN based image steganography technique with a new type of attention mechanism that can better capture detailed textures; therefore, it is more imperceptible and has enhanced security for hidden data.
- 2) We introduce a Residual-Spatial Attention (RSA) block to capture richer local features through multiple scales of skip connections within consecutive convolutional layers.
- 3) Moreover, in terms of optimisation for the DenseNet network, we enhanced its depth while improving tex-

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ture capture capacity; henceforth, this can increase the model's capabilities to learn features.

II. RELATED WORK

Invertible Neural Networks (INNs) have been effectively utilized in diverse image processing applications, including image denoising, dehazing, and generation. A key feature of INNs is their use of identical network parameters for both the encoding and decoding stages, which facilitates lossless information processing and ensures the preservation of fine input data details. Recently, the potential of these networks has been extended to image steganography, where the embedding and extraction of images can be conceptualized as inverse processes.

Several methods, such as DeepMIH [5], Hinet [6], and RIIS [7], employ the bidirectional information flow of INNs to perform image steganography. A major limitation of these approaches, however, is their dependence on traditional coupling and decoupling layers. Stego images produced under this paradigm tend to sustain unintended edge characteristics inherited from the secret image, posing a risk to steganographic security. Such an artifact not only compromises the visual similarity between the cover and stego images but also creates significant security vulnerabilities, particularly in situations where the original cover image might be available to an adversary.

The attention mechanism presents a potential solution by enabling a network to concentrate on salient image regions while ignoring irrelevant perceptual details, thereby offering superior guidance for identifying the best locations for embedding secret data. Currently, reconciling the inherent trade-offs among embedding robustness, image quality, and capacity in complex multi-image scenarios has proven difficult for existing methodologies. Consequently, this study explores the synergy of attention mechanisms within a steganographic framework to offer superior strategic guidance for concealing secret information.

III. PROPOSED APPROACH

This section presents a reversible hiding-revealing framework, which is intended to facilitate image steganography with large capacity, improved security, and high imperceptibility.

A. Overall structure

Figure 1 illustrates the architecture of our proposed invertible steganographic network, which we have named RSAINN. This efficient, end-to-end deep invertible network is specifically designed for steganography and features the integration of a Residual-Spatial Attention (RSA) mechanism. The architecture primarily consists of Discrete Wavelet Transform (DWT), Inverse Wavelet Transform (IWT), and multiple invertible network blocks. To begin the forward hiding stage, x_{cover} and x_{secret} are fed into the system. These input images undergo DWT decomposition first. This step partitions them into various low-frequency and high-frequency wavelet subbands for further processing. The resulting feature maps

are then processed by a series of invertible hiding blocks, which incorporate our RSA module to enhance the extraction of salient image information. Finally, the IWT is applied to reconstruct these features, yielding the stego image x_{stego} . This output encapsulates the embedded secret image along with auxiliary information r , which accounts for any potential data loss during the transformation. For the reverse revealing process, the inverse network receives the stego image x_{stego} and an auxiliary random variable z as input. Following a similar procedure, these inputs undergo a DWT and are processed by the revealing blocks, which also leverage the RSA module. Subsequently, the features are processed by the IWT. This step ensures that the hidden image is restored accurately.

B. DWT/IWT module

Pixel-domain embedding often causes detectable artifacts like texture replication and color distortion. We hide information in the frequency domain to improve both security and robustness. In this approach, the secret image is embedded into the high-frequency parts of the cover image. Specifically, we utilize the Discrete Wavelet Transform (DWT) due to its perfect reconstruction and mathematical rigor. The DWT module decomposes the cover image from dimensions (B, C, H, W) into subbands of $(B, 4C, H/2, W/2)$, where B, C, H, W denote batch size, channels, height, and width, respectively. As complementary inverse operations, DWT and the Inverse Wavelet Transform (IWT) ensure lossless end-to-end optimization. We implement the Haar wavelet for its efficiency. We define the forward and reverse steganographic procedures mathematically. The related formulations for concealment and extraction are shown in Eq. 1 and Eq. 2:

$$\begin{aligned} (x_{cover}, x_{secret}) &= \text{DWT}(X_{cover}, X_{secret}) \\ (X_{stego}, r) &= \text{IWT}(x_{stego}, R) \end{aligned} \quad (1)$$

$$\begin{aligned} (x_{stego}, Z) &= \text{DWT}(X_{stego}, z) \\ (X_{cov_rev}, X_{sec_rev}) &= \text{IWT}(x_{cov_rev}, x_{sec_rev}) \end{aligned} \quad (2)$$

C. Residual-Spatial Attention Module

In order to improve the boundary's characteristic acquisition ability of traditional DenseNet, a novel Residual-Spatial Attention (RSA) architecture integrating residual learning with spatial attention modules is proposed herein. As shown in Figure 2, the RSA structure consists of several convolutional (Conv) units and ReLU activation functions connected together. Using dense skip connections and element-wise addition functions to integrate many of the local features obtained by subsequent convolutional layers.

But using a single layer of convolution fails to enhance the scope of observation beyond this area because of its limited range of view. To solve the above problem, add an attention mechanism to improve the modelling ability for non-local features. By integrating strong reinforcement of Skips and attention-assisted Feature refinement in RSA modules to improve robustness of features extraction and multi-level fusion, it has achieved outstanding results on image steganography tasks.

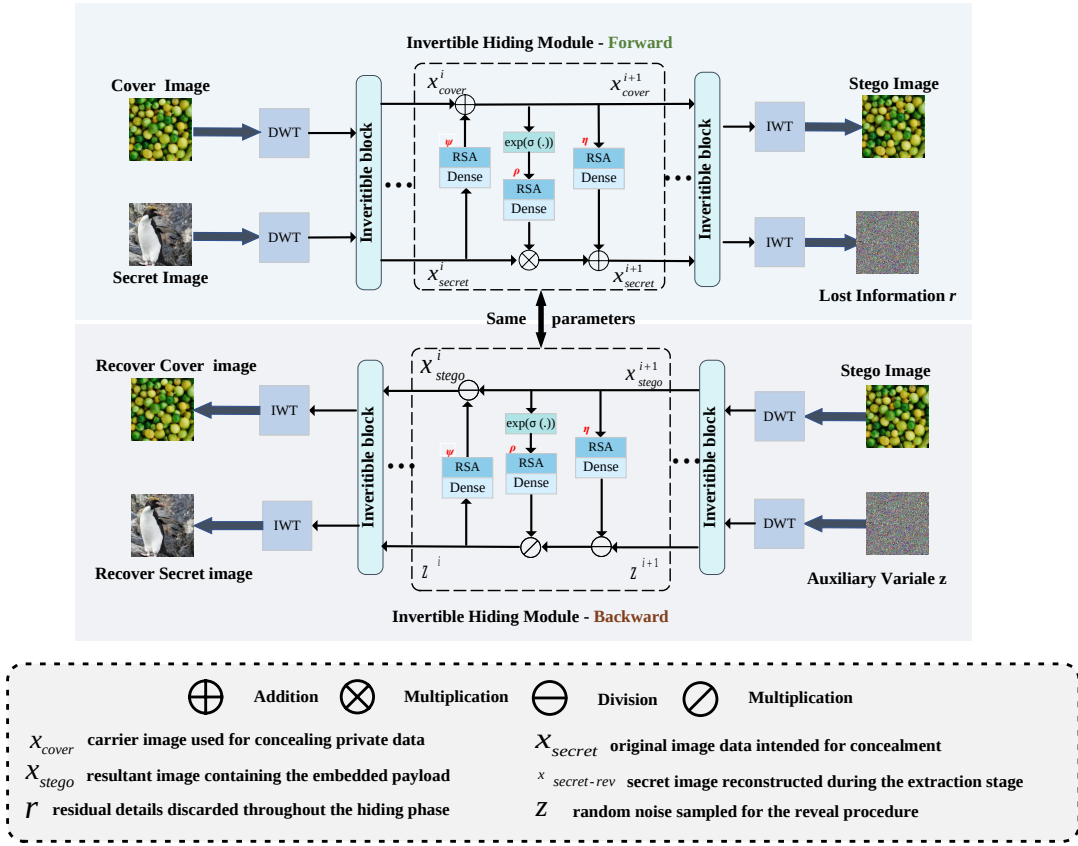


Fig. 1. Overall structure of the proposed RSAINN framework

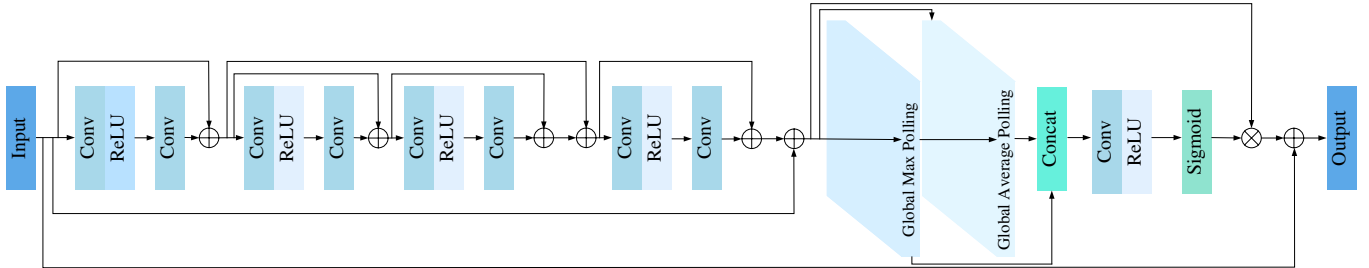


Fig. 2. Components and structure of the Residual-Spatial Attention

D. Invertible hiding module

Based on this, a stack structure of several invertible modules was designed in the following way to enhance learning ability. Particularly, in terms of the convolutional part at each block, an improvement is made that enhances its compatibility with the whole system and improves detail extraction; At this time, it must ensure invertibility.

In order to enhance feature acquisition, inspired by the DenseNet structure for building a convolutional network with Residual-Spatial Attention (RSA) as its core component inside it. To enable stronger expression of features through combination with dense connectivity and attention-driven fine-tuning.

A certain number of parameter values are reused by the

forward-embedding phase of the secret image x_{secret} and the reverse recovery procedure for recovering the stego image x_{stego} . such a parameter sharing can maintain structural integrity and enhance optimisation performance of both directions.

As illustrated in Figure 1, during the i -th forward hiding stage, the invertible block takes x_{cover}^i and x_{secret}^i as inputs and produces the corresponding outputs x_{cover}^{i+1} and x_{secret}^{i+1} through the embedding operation. The overall transformation process

can be mathematically formulated as shown in Eq. 3.

$$\begin{aligned} x_{\text{cover}}^{i+1} &= x_{\text{cover}}^i + \psi(x_{\text{secret}}^i) \\ x_{\text{secret}}^{i+1} &= x_{\text{secret}}^i \cdot \exp(\alpha(\rho(x_{\text{cover}}^{i+1}))) \\ &\quad + \eta(x_{\text{cover}}^{i+1}) \end{aligned} \quad (3)$$

Where, $\psi(\cdot)$, $\rho(\cdot)$, and $\eta(\cdot)$ represent differentiable transformation functions. In common practice, these are based on dense blocks following [8]. Our design uses the RSADense block instead. This change adds attention mechanisms to the transformation layers. The symbol η indicates the sigmoid activation, and $\cdot$ stands for element-wise multiplication. After the last invertible block, the network outputs x_{cover}^M and x_{secret}^M . These are then converted by the IWT into the stego image X_{stego} and residual r .

The extraction process works in the opposite direction but shares the same parameters. At the $(i+1)$ -th stage, x_{stego}^{i+1} and z^{i+1} are input into the invertible block. These DWT-transformed components are processed to yield x_{stego}^i and z^i . The specific mathematical steps for this reverse operation are shown in Eq. 4:

$$\begin{aligned} x_{\text{stego}}^i &= x_{\text{stego}}^{i+1} - \psi(z^i), \\ z^i &= \frac{\eta(x_{\text{stego}}^{i+1}) - z^{i+1}}{\exp(\alpha(\rho(x_{\text{stego}}^{i+1})))} \end{aligned} \quad (4)$$

In this process, z is sampled from a Gaussian distribution. After M iterations, the network produces x_{stego}^1 . The secret image $X_{\text{sec_rev}}$ is then reconstructed using the IWT. This final step completes the recovery of the hidden information.

E. Integration of Residual-Spatial Attention within DenseNet

Our invertible neural network uses a densely connected topology for its foundation. Specifically, this backbone supports the transformation functions $\psi(\cdot)$, $\rho(\cdot)$, and $\eta(\cdot)$. In contrast to conventional designs that tend to learn limited feature representations, we enhance the expressive capacity of the model by embedding the RSA into the dense architecture, forming the DenseNet framework.

Figure 3 illustrates the structure of this network, the reversible block is then enhanced by adding more than one additional layer of convolution to reach six. Finally, using these feature maps, we will proceed to refine them through the RSA module. This design can promote better feature Propagation and reuse at different levels.

In particular, the RSA mechanism strengthens feature extraction through two parallel operations of adaptive channel-weighting and spatial attention assignment. When jointly modelling dependencies in channels and weights to extract more discriminative features from networks, such improvements have enhanced performance for some downstream steganography applications.

IV. EXPERIMENTS

Experimental verification of the performance of the proposed RSAINN is conducted in this part. Tested in multiple

large-scale image datasets and compared with a group of recently published state-of-the-art (SOTA) multi-image steganographic systems. In all experiment settings, please refer to below:

A. Experimental setup

1) *Datasets*: Experiments are carried out on three benchmark datasets, namely DIV2K, COCO, and ImageNet. For data preprocessing, images from DIV2K are center-cropped to a resolution of 1024×1024 . For COCO and ImageNet, 5,000 and 50,000 images are randomly selected, respectively, and resized to 256×256 . We split each dataset into three parts: training, validation, and testing, using an 8:1:1 distribution.

2) *Experimental environment*: Three widely used benchmarks of image data sets, including DIV2K, COCO and ImageNet, respectively. Data pre-processing: Centre crop the images in DIV2K at a size of 1024×1024 . For COCO and ImageNet, 5000 and 50000 images were randomly chosen, then rescaled to size 256×256 each. Each set of data is divided among training, validation and testing in a proportion of 8:1:1. During the process of training with dynamic random cropping, whereas test times use central crop processing as standard size alignment for comparisons.

3) *Quantitative evaluation*: The performance of our approach is assessed using four standard indicators. These include PSNR and SSIM for quality analysis, alongside RMSE and MAE for error estimation. This combination provides a comprehensive view of the steganographic results. Furthermore, to verify the security of this technique using well-known stegananalytical software programs StegExpose and SRNet, respectively. Among them, StegExpose applies the method of statistic-based feature analysis; SRNet adopts learnt-driving-detection model technology.

B. Experimental comparison and analysis

1) *Comparison with other methods*: To validate RSAINN, we compared it with seven SOTA methods. For fairness and reproducibility, all methods were retrained in a unified environment; thus, results may slightly differ from original reports. Table I shows the results.

Table I presents a comparison of secret image recovery performance between the proposed method and several recent works on three datasets. As shown in the results, our method consistently achieves superior performance in secret image reconstruction. In terms of PSNR, our method outperforms the best competing approach by 1.8 dB. Specifically, across the three datasets, the improvements are 6.61 dB, 4.8 dB, and 3.97 dB, respectively, demonstrating consistently better performance than other methods in all cases.

The above improvements can be attributed to the effectiveness of the proposed framework, which integrates residual and spatial attention modules. By combining channel-wise importance and spatial dependencies, the network achieves more informative feature representations while suppressing noise interference. The residual component provides noise reduction capability, whereas the spatial attention mechanism

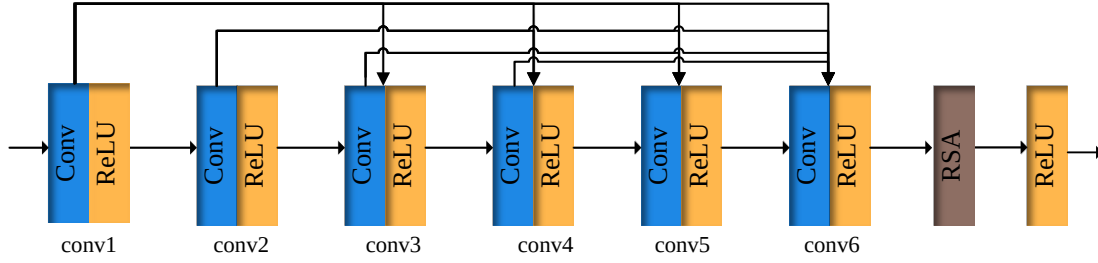


Fig. 3. Integration of RSA modules within the DenseNet architecture

TABLE I
DIFFERENT METHODS' COMPARISON ACROSS DIVERSE IMAGES AND DATASETS.

Recovery/Secret Image												
Methods	DIV2K				COCO				ImageNet			
	PSNR	SSIM	MSE	RMSE	PSNR	SSIM	MSE	RMSE	PSNR	SSIM	MSE	RMSE
LSB [9]	35.03	0.9611	5.30	6.49	34.82	0.9561	5.36	6.55	34.91	0.9568	5.35	6.53
Baluja [10]	37.25	0.9587	5.73	7.65	36.99	0.9575	5.83	7.72	36.71	0.9578	6.22	8.17
Weng et al. [11]	39.57	0.9788	3.08	4.25	36.73	0.9672	4.06	5.78	36.87	0.9668	4.13	5.79
HiNet [12]	48.82	0.9959	0.72	1.05	41.81	0.9859	1.51	2.37	41.93	0.9837	1.61	2.44
DeepMIH [13]	43.72	0.9895	1.94	2.81	40.30	0.9805	2.83	4.14	40.31	0.9800	2.87	4.16
iSCMIS [14]	45.78	0.9924	1.62	2.42	41.53	0.9818	2.53	3.78	41.64	0.9818	2.59	3.79
cINN [15]	46.47	0.9985	2.06	2.35	43.26	0.9886	2.28	2.38	46.27	0.9914	2.12	2.48
Ours	49.17	0.9969	0.89	0.92	49.87	0.9974	1.02	0.91	51.07	0.9977	0.91	0.83

Stego/Cover Image												
Methods	DIV2K				COCO				ImageNet			
	PSNR	SSIM	MSE	RMSE	PSNR	SSIM	MSE	RMSE	PSNR	SSIM	MSE	RMSE
LSB [9]	24.99	0.8951	15.57	18.16	24.96	0.8939	15.31	17.93	25.00	0.8960	15.27	17.90
Baluja [10]	36.34	0.9716	4.47	6.01	34.85	0.9558	5.33	7.34	34.82	0.9547	5.43	7.44
Weng et al. [11]	37.43	0.9487	3.58	4.96	35.13	0.9475	4.66	6.70	35.04	0.9444	4.81	6.86
HiNet [12]	42.74	0.9861	1.23	1.81	38.37	0.9682	2.24	3.36	38.53	0.9662	2.33	3.39
DeepMIH [13]	41.41	0.9801	2.35	3.43	36.55	0.9613	4.09	6.04	36.63	0.9604	4.16	6.07
iSCMIS [14]	42.53	0.9836	2.11	3.04	37.48	0.9664	3.74	5.48	37.69	0.9661	3.79	5.48
cINN [15]	42.63	0.9851	2.15	2.88	41.35	0.9824	2.69	3.13	40.87	0.9766	2.71	3.25
Ours	42.84	0.9818	3.67	1.88	43.71	0.9667	2.08	1.71	44.52	0.9628	2.64	1.57

enhances information distribution and improves robustness, thereby achieving a better balance between detail preservation and steganographic security in terms of reconstruction performance. In terms of visual quality, both cover and corresponding steganographic images demonstrate superior reconstruction performance compared with other methods. The PSNR values achieved by the proposed method are consistently higher than those of competing approaches across all three datasets, namely DIV2K, COCO, and ImageNet. The improvements for each dataset are 1 dB, 2.36 dB, and 3.65 dB, respectively. On the DIV2K dataset, the MSE and RMSE values are slightly higher than those of other state-of-the-art methods. This is mainly because the training objective places greater emphasis on PSNR optimization, which is prioritized over MSE and RMSE losses. However, this trade-off is acceptable considering the overall performance improvement. This minor compromise, the overall performance remains advantageous when considering multiple evaluation metrics.

2) *Histogram analysis*: Histogram analysis frequently describes images' properties through examination of the distribution of their pixels' intensity values. Due to the different distributions of each image, even if there is some deviation in pixels' values, it will affect all histograms similarly. Therefore, comparing the histograms of a processed image and its original form is a reliable method to evaluate whether secret information has been concealed after training.

Among them, four typical pictures were randomly chosen for histogram analysis. As shown in Figure 4, the distribution of the cover image and its corresponding stego-image is very similar. Visually, there is no difference among them; thus, the reconstructed images of these areas extracted will still be similar to each other. There was no significant deviation from the pixel distribution before embedding in this study's results. According to the above conclusions, our method simultaneously possesses sufficient imperceptibility and a high-reliability information-embedding capacity.

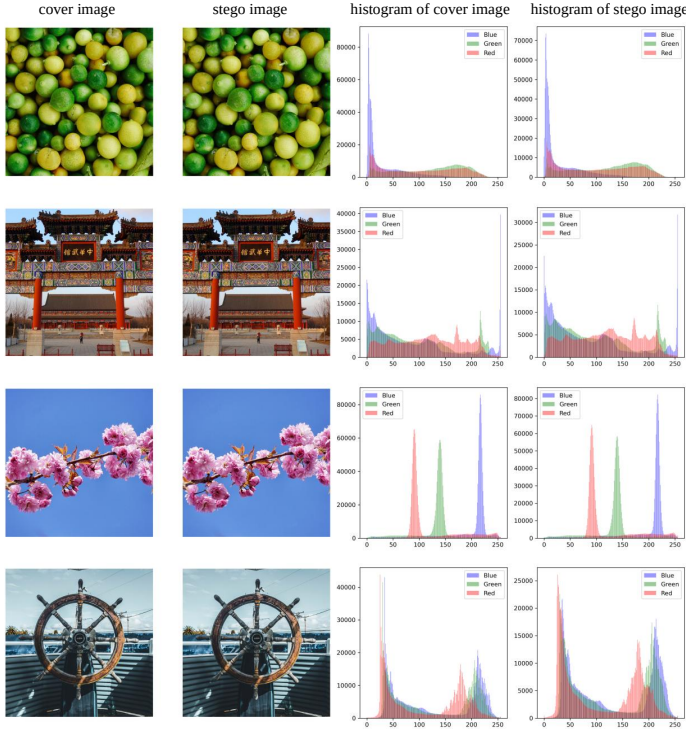


Fig. 4. Comparison of image statistical characteristics

3) *Steganographic analysis*: Steganalysis can be used to determine whether images have been tampered with by evaluating the detection capability of hidden information embedded within them. In particular, this paper aims to identify which stego images are more distinguishable from their corresponding cover images using specific detection methods. Two evaluation approaches, StegExpose and SRNet, are employed in this study; one is based on traditional statistical analysis, while the other is trained using a deep learning algorithm.

a) *Statistical steganalysis*: We first utilize StegExpose to perform statistical analysis of the generated stego images. A total of 2,500 cover–stego image pairs are randomly sampled from the test set for evaluation. By varying the decision threshold from 0 to 1, the Receiver Operating Characteristic (ROC) curve is obtained, as illustrated in Figure 5.

The results indicate that StegExpose exhibits limited capability in distinguishing the stego images produced by our method. The Area Under the Curve (AUC) reaches 0.5182, which is very close to 0.5, corresponding to random-level performance. This suggests that the statistical traces introduced by the embedding process are minimal, thereby providing strong resistance against conventional steganalysis techniques.

b) *Deep learning-based steganalysis*: Firstly, we use StegExpose for statistical analysis of the generated stego images. A total of 2,500 cover–stego image pairs are randomly selected from the test set for evaluation. By varying the decision threshold from 0 to 1, the Receiver Operating Characteristic (ROC) curve is obtained, as shown in Figure 5.

As shown in the results, StegExpose has relatively weak

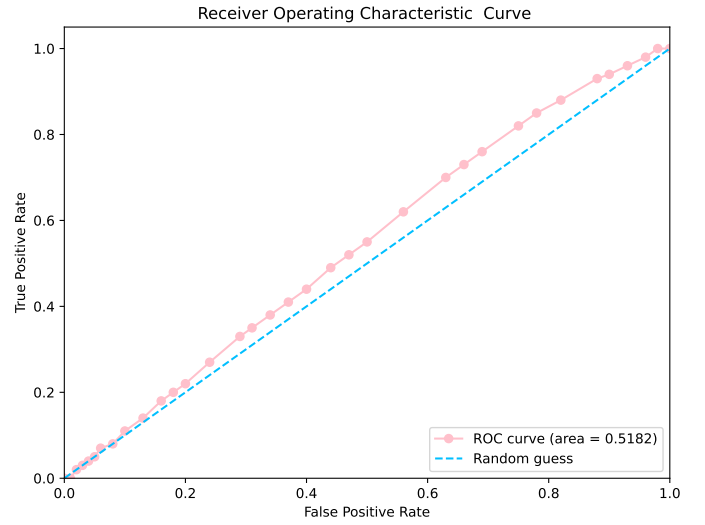


Fig. 5. Steganalysis results against the StegExpose tool

capability in detecting stego images generated by the proposed method. The Area Under the Curve (AUC) is approximately 0.5182, which is very close to 0.5, indicating random-level performance. Therefore, it can be inferred that the statistical artifacts introduced by the embedding process are minimal and difficult to detect using traditional steganalysis methods.

TABLE II
DETECTION ACCURACY RATES UNDER SRNET STEGANALYSIS

Method	Accuracy (%)
LSB	99.13 $\pm$ 0.43
Baluja	99.58 $\pm$ 0.21
Weng et al.	77.43 $\pm$ 2.01
HiNet	55.86 $\pm$ 0.27
DeepMIH	75.54 $\pm$ 0.67
iSCMIS	69.64 $\pm$ 1.38
cINN	53.71 $\pm$ 0.34
Ours	50.83 $\pm$ 0.74

4) *Ablation Study*: We conduct ablation experiments to evaluate the impact of the Residual Attention (ResA) module, Spatial Attention (SpaA), and optimized DenseNet blocks. The results summarized in Table III indicate that our complete model offers a clear advantage over the baseline. Specifically, the recovery PSNR for secret images shows an increase of 9.03 dB. Furthermore, we observe a 12.4% improvement in steganographic quality compared to the baseline. The results indicate that skip connections help preserve fine-grained details, while spatial attention focuses on semantically important regions. When combined in the RSA module, these components exhibit a strong synergistic effect, leading to substantial performance improvements. This integration enables both local feature preservation and global context awareness. In addition, the reversibility of the framework ensures stable information recovery, further demonstrating the feasibility and complementarity of the proposed design.

TABLE III
ABLATION STUDY

ResA	SpaA	RSA	DenseNet	Recovery/Secret		Stego/Cover	
				PSNR	SSIM	PSNR	SSIM
✓	✓	✓	✓	41.57	0.9824	31.44	0.8289
				43.25	0.9868	35.82	0.9125
				44.63	0.9892	37.45	0.9347
	✓	✓	✓	45.59	0.9927	40.53	0.9519
				45.43	0.9916	40.26	0.9614
		✓	✓	49.17	0.9969	42.84	0.9818

V. CONCLUSION

This research leverages the reversibility of Invertible Neural Networks to ensure lossless secret information extraction. Our Residual-Spatial Attention (RSA) module captures global and contextual dependencies, enabling the network to adaptively embed secret data into optimal cover regions. This design effectively balances visual fidelity and security, even at high embedding capacities. Comparative experiments demonstrate that our methodology outperforms existing methods in visual quality and robustness across multiple datasets. However, the model's complexity demands significant computational resources. Future work will focus on enhancing security for practical deployment and exploring lightweight architectures for resource-constrained edge environments.

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